

# Vol 13 Chapter 5 — Evolution as Substrate Dynamics

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

## Chapter 5 — Evolution as Substrate Dynamics

*Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine\_DIV5\_QM\_GR\_SM/) or the theorem registry disagree, they win.*

### Level 1 — one sentence

Darwinian evolution — variation (mutation) + heredity (DNA replication) + selection (differential survival/reproduction) → adaptation — is the population-level dynamics of substrate K-type configurations across biological time, with mutations as substrate K-type alterations and selection as substrate-permeated environmental filtering.

### Level 2 — graduate-physicist precision

#### 5.1 Darwin's mechanism

Darwin 1859, *Origin of Species*: natural selection operates on heritable variation. Modern synthesis (1930s-40s, Fisher-Haldane-Wright): mathematical population genetics integrates Mendel + Darwin.

Selection equation (continuous, single locus):

$$\frac{dp}{dt} = sp(1 - p)$$

with  $p$  allele frequency,  $s$  selection coefficient. Fixation time  $\sim 1/s$  generations for positive  $s$ .

## 5.2 Mutation rates

Per-base-pair per generation: - Bacteria (*E. coli*):  $\sim 10^{-9}$  - Mammals:  $\sim 10^{-8}$  - Viruses (RNA):  $\sim 10^{-4}$

Genome-wide:  $\sim 1$  mutation per replication for bacteria,  $\sim 100$  for mammals.

## 5.3 Genetic drift

Random sampling in finite populations: drift dominates when  $|s| < 1/(2N_e)$  with  $N_e$  effective population size.

Neutral theory (Kimura): most molecular evolution is drift, not selection. Molecular clock = drift accumulating mutations.

## 5.4 Substrate dynamics view

Substrate provides: - Mutation: substrate K-type alteration (radiation, chemical damage, replication error) - Selection: substrate-mediated environment filters K-type configurations - Drift: substrate stochasticity at sampling

BST observation: substrate dynamics include both deterministic (selection on K-type stability) and stochastic (drift, mutation) components — same epistemic-stochasticity reading as DCCP (memory).

## 5.5 Phylogenetics

Reconstruct evolutionary history from molecular sequences. Tree-of-life topology: 3 domains (Bacteria, Archaea, Eukarya) with horizontal gene transfer making the tree more of a web.

Common ancestor: LUCA  $\sim 3.5$  Gya. Eukaryotes  $\sim 1.8$  Gya. Animals  $\sim 700$  Mya. Vertebrates  $\sim 500$  Mya. Primates  $\sim 85$  Mya. *Homo sapiens*  $\sim 300$  Kya.

## 5.6 K-audit anchors

- Vol 6 Ch 1-4 (statistical mechanics  $\leftrightarrow$  population genetics)
- Vol 14 (information theory  $\leftrightarrow$  evolution)
- DCCP (memory; epistemic-deterministic substrate dynamics)

## Level 3 — 5th-grader accessibility

Evolution = variation + heredity + selection  $\rightarrow$  adaptation. Darwin 1859. Modern math:  $dp/dt = sp(1 - p)$ . Mutation rate:  $10^{-9}$  to  $10^{-4}$  per base per generation. Drift vs selection: stochastic vs deterministic. In BST: mutations are substrate K-type alterations; selection is substrate-environmental filtering; same epistemic-stochasticity as DCCP. LUCA  $\sim 3.5$  Gya. Humans  $\sim 300$  Kya.

What comes next

Chapter 6 develops biochemistry bridge to Vol 12 Chemistry.

Where to look this up

- Darwin 1859
- Fisher-Haldane-Wright (Modern Synthesis)
- BST: Vol 6, Vol 14, DCCP